

Eden Danielsen–Jensen

Ph.D. Candidate in Mathematics

Department of Mathematics, Brigham Young University

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Education

Brigham Young University

Ph.D. in Mathematics

Expected Graduation: August 2027

- Advisor: Jennifer Brooks.
- Dissertation: Group-invariant CR Maps from Spheres to Hyperquadrics (working title).
Investigates the possible target dimensions and signatures of group-invariant CR maps from spheres in $\mathbb{C}^2$ to hyperquadrics, including constructions and regions where no such maps exist.

University of the Sunshine Coast

Bachelor of Science (Honours), Mathematics — First Class Honours

2022

- Advisors: Robert McDougall & Lauren Thornton.
- Honours thesis: *A Revised Introduction for the Base Semisimple Class*.
Used class operators to reinterpret existing results in radical theory concerning the base-semisimple class.

University of the Sunshine Coast

Bachelor of Science — GPA 6.96/7.00

2021

- Major: Mathematics
- Minor: Chemistry

Research Interests

Several Complex Variables, CR Geometry, proper holomorphic mappings, group-invariant CR maps, hyperquadrics, invariant polynomial maps, computational methods in complex analysis.

Publications

In Preparation

- Jennifer Brooks, Eden Danielsen–Jensen, Dusty Grundmeier, and Alex Lee. *Lower Bounds for Invariant CR Maps*. In preparation.
- Eden Danielsen–Jensen, Dusty Grundmeier, Abdullah Al Helal, Valentin Kunz, Jiří Lebl, Ming Xiao, and Weixia Zhu. *Degree-Three Rational Sphere Maps: Sharp Denominator Region and Gram Normal Forms*. [arXiv:2608.29451](https://arxiv.org/abs/2608.29451).

Presentations

Signatures for Group-Invariant Monomial Maps from Spheres to Hyperquadrics

- Joint Mathematics Meetings (JMM), Washington, DC January 2026
- MAA Intermountain Section Meeting, Provo, UT March 2026
- BYU Student Research Conference, Provo, UT February 2026

A Revised Introduction for the Base Semisimple Class

- Australian Mathematical Society Conference, Sydney, NSW December 2022
- Australian Algebra Conference, Sunshine Coast, QLD December 2022

Research Activities

Collaborative Research Workshop in Several Complex Variables

Institute for Advanced Study (IAS), Princeton NJ

June 2026

Teaching Experience

Instructor of Record — Brigham Young University

- **MATH 213** Elementary Linear Algebra Summer 2026
- **MATH 109** Precalculus Winter 2026
- **MATH 116** Essentials of Calculus Fall 2025
- **MATH 112** Calculus I Spring 2025

Teaching Assistant / Recitation Leader — Brigham Young University

- **MATH 341** Theory of Analysis I Fall 2025, Fall 2026
- **MATH 113** Calculus II Fall 2026

Grader — Brigham Young University

- **MATH 570** Matrix Analysis Winter 2025
- **MATH 510** Numerical Linear Algebra Fall 2025
- **MATH 314** Calculus of Several Variables Summer 2024

Sessional Academic Tutor — University of the Sunshine Coast

- **MATH 104** Introductory Calculus Semester 2 2023
- **MATH 103** Introduction to Applied Mathematics Semester 1 2023
- **SCI 110** Science Research Methods Semester 1 & 2 2023

Service & Outreach

- Graduate Student Cohort Representative, BYU Mathematics 2024–Present
- Organizer, Graduate Student Writing Group, BYU Mathematics 2026–Present
- Organizer, Graduate Professional Development Workshops, BYU Mathematics 2025–Present
- Graduate School Q&A Panelist, BYU Mathematics 2025, 2026
- Creator and Facilitator, Mathematical Support Sessions, UniSC 2023
- Conference Organization Support, Australian Algebra Conference 2022

Honors & Awards

- Vice-Chancellor's Honours Scholarship, University of the Sunshine Coast 2022
- Thompson Excellence Scholarship, University of the Sunshine Coast 2019–2021
- Rural and Regional Enterprise Scholarship 2019–2021

Professional Memberships

- American Mathematical Society Member
- Mathematical Association of America Member
- Australian Mathematical Society Member
- Women in Mathematics Special Interest Group (WIMSIG) AustMS Member